25900146 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCMOS100 - FOUNDATIONS OF MOBILE DEVELOPMENT SYSTEMS

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. How do you remove an element from a List? [U] / [CO1]

   a). list.delete(value); b). list.remove(value);

   c). list.pop(value); d). list.clear(value);

2. Which of the following control flow statements is used to terminate a loop? [U] / [CO1]

   a). continue b). stop c). break d). exit

3. What is encapsulation? [U] / [CO1]

   a). Hiding the details and showing only the necessary parts. b). Creating objects from a class

   c). Using methods only. d). Inheriting from other classes.

4. What is the first step in setting up flutter on a new machine? [K] / [CO2]

   a). Installing flutter SDK b). Creating a new project c). Running flutter doctor d). Configuring the IDE

5. What is the advantage of using adaptive widgets in Flutter? [U] / [CO2]

a). They automatically adjust their appearance based on the platform.
b). They reduce the need for state management.
c). They prevent UI updates in a Stateful widget.
d). They replace the need for setState().

6. Which of the following does a Asset folder does not contain ? [K] / [CO2]

a). Images
b). Fonts
c). static data
d). Configuration files

7. What is the primary purpose of a GET request in a RESTful API? [K] / [CO3]

a). To create a new resource
b). To update an existing resource
c). To retrieve data from a server
d). To delete a resource

8. Which storage option provides real-time database synchronization in Flutter? [U] / [CO3]

a). SQLite
b). Cloud Firestore
c). SharedPreferences
d). Hive

9. Which of the following is used to persist data locally using key-value pairs in Flutter? [K] / [CO3]

a). Cloud Firestore
b). SQLite
c). SharedPreferences
d). Firebase Authentication

10. What does Firestore allow you to do in real-time? [U] / [CO3]

a). Synchronize files between devices
b). Synchronize data changes across devices
c). Store encrypted data
d). Track user activities

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Explain the difference between Set and List in Dart. [U] / [CO1]

12. What is exception handling, and why is it important in programming? Explain with an example. [K] / [CO1]

13. How does Shared Preferences store data in a Flutter application? [U] / [CO2]

14. What is the key difference between a Stateless and a Stateful widget in Flutter? [K] / [CO2]

15. What are the differences between Firebase Authentication and Firebase Firestore in Flutter? [U] / [CO3]

16. What is the purpose of the DELETE method in RESTful APIs? [K] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Discuss the advantages and disadvantages of using Lists, Sets, and Maps in Dart. [K] / [CO1]

18. Explain the concept of classes in Dart. How do they help in organizing and structuring Flutter applications? [K] / [CO1]

19. Explain the role of setState() in Flutter state management. How does it impact UI updates? Provide examples. [U] / [CO2]

20. How does Flutter's reactive UI system differ from traditional imperative UI frameworks? [U] / [CO3]

[10x2 = 20]